TT-T-662b
JULY 6, 1960
SUPERSEDING
Int. Fed. Spec. TT-T-00662a(GSA-FSS)
October 26, 1959, and
Fed. Spec. TT-T-662
December 15, 1949

FEDERAL SPECIFICATION

TRIPHENYL PHOSPHATE (FOR USE IN ORGANIC COATINGS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers triphenyl phosphate satisfactory for use in organic coatings.

1.2 Classification.

1.2.1 *Type and grade.*—This specification covers one grade and one type of triphenyl phosphate.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

U-M-186—Medicinal-Products and Clinical-Laboratory Reagents; General Specifications for Containers (Packaging and Packing).

PPP-B-636—Box, Fiberboard.

PPP-C-96—Cans, Metal, 28 Gage and Lighter.

PPP-D-729 — Drums: Metal, 55-Gallon (For Shipment of Noncorrosive Material).

PPP-D-760 — Drums and Pails, Metal (15 and 16.64 Gallon).

PPP-T-60 — Tape: Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver,

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San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–C–3955 — Cans Fiber, Spirally-Wound.

MIL–T–4239 — Tape, Pressure-Sensitive Adhesive, Vinyl-Plastic, Opaque, Photographic.

Military Standards:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Pharmacopoeia of the United States (15th Edition).

(The Pharmacopoeia of the United States is distributed by the Mack Publishing Company, Easton, Pa.)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Quantitative requirements. — The triphenyl phosphate shall conform to the requirements of table I.

3.2 Qualitative requirements.

3.2.1 *Appearance.*—The appearance of the triphenyl phosphate shall be that of white flakes.

3.2.2 *Odor.* — The material shall have a faintly aromatic odor, characteristic of triphenyl phosphate.

TABLE I.—*Quantitative requirements*

Characteristics	Requirements	
	Minimum	Maximum
Melting point °C.	48.5	---
Nonvolatile matter, percent by weight	99.8	---
Acidity, mg. of KOH per gram of sample	---	0.10
Ester content, percent by weight as triphenyl phosphate	99	---
Free phenols, percent by weight	---	0.15

3.2.3 *Oxidizable substances.* — The triphenyl phosphate shall be free from oxidizable substances, when examined as described in paragraph 4.3.6.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling and inspection.—Unless otherwise specified, sampling and inspection shall be in accordance with the applicable methods 1011, 1021 and 1031 of Federal Test Method Standard 141.

4.3 Test procedures.—Tests shall be made in accordance with the applicable methods of Federal Test Method Standard No. 141.

4.3.1 The following tests shall be conducted in accordance with Federal Test Method Standard No. 141.

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Test	Method
Nonvolatile matter	[1]4041

[1] Use porcelain dish instead of tin lid.

4.3.2 *Melting point.*—Determine the melting point by the capillary tube method included in the U. S. Pharmacopoeia (15th Edition).

4.3.3 *Acidity.* — Place approximately 25 grams of the sample into a 200 milliliter Erlenmeyer flask containing 50 milliliters of ethyl alcohol previously neutralized using thymol blue as indicator. Dissolve and titrate with 0.1*N* potassium hydroxide in 99 percent ethyl alcohol. From the weight of the sample taken and the volume of alcoholic potassium hydroxide used, calculate the number of milligrams of potassium hydroxide required for the neutralization of 1 gram of the sample.

4.3.4 *Ester content.*—Accurately weigh 1 to 2 grams of the sample into a 200-milliliter Erlenmeyer flask containing one and one-half times the theoretical quantity of 0.5 *N* alcoholic potassium hydroxide required for complete saponification. Attach a reflex condenser to the flask and heat on a steam bath until saponification is complete (8 to 16 hours). During the heating, shake the set-up and contents frequently, taking the usual precautions to lose none of the contents. Cool, wash down the condenser with distilled water, remove the condenser, and add three drops of phenolphthalein indicator to the contents of the flask. Titrate the contents of the flask with 0.5 *N* hydrochloric acid. Two blanks of the alcoholic potassium hydroxide shall be run along with the sample. The blanks shall check within one-tenth of a milliliter. Calculate the percentage of ester by weight, allowing in the calculation for acidity as determined in paragraph 4.3.3.

(Factor — 1 ml. of 0.5 *N* KOH = 0.0544 gram of triphenyl phosphate.)

4.3.5 *Free phenols.*

4.3.5.1 *Reagents.*

1. A 2.5 percent aqueous sodium hydroxide solution.
2. Sodium bromide-bromate solution containing 10.09 grams of sodium bromide and 2.95 grams of sodium bromate per liter of distilled water.
3. A 6-percent aqueous solution of potassium iodide.
4. Hydrochloric acid (sp. gr. 1.19).
5. A 1-percent starch solution.
6. A 0.05 *N* solution of sodium thiosulfate.

4.3.5.2 *Procedure.* — Weigh approximately 10 grams of the sample to the nearest milligram into a 250-milliliter volumetric flask. Add 50 milliliters of the sodium hydroxide solution warmed to 65°C. and shake the flask vigorously for 3 minutes. Dilute with distilled water to the 250-milliliter mark and filter. Pipette 100 milliliters of the filtrate into a 300-milliliter flask and add from a burette 15 milliliters of the sodium bromide-bromate solution. Immediately add 5 milliliters of the hydrochloric acid, stopper the flask, thoroughly mix its contents by shaking and allow to stand for 15 minutes. Add 10 milliliters of the potassium iodide solution and titrate with the sodium thiosulfate solution, using starch solution as the indicator. Add the starch solution near the end of the titration. The fading out of the blue color marks the end point of the titration. Run a blank titration as described above using the same reagents.

4.3.5.3 *Calculation.*—Calculate the percentage of free phenols as follows:

$$C = \frac{(a - b) \times N \times 4.5}{W}$$

Where:

C = Percent of free phenols.
a = Milliliters of $Na_2S_2O_3$ for titration of blank.

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b = Milliliters of $Na_2S_2O_3$ for titration of sample.
N = Normality of $Na_2S_2O_3$.
W = Weight of sample.

4.3.6 *Oxidizable substances.*—Shake vigorously for 2 minutes a 10-gram sample with 50 milliliters of freshly prepared 0.01 N potassium permanganate solution. The purple color shall remain unchanged to 30 minutes.

4.3.7 *Appearance.* — Examine the sample for conformance to paragraph 3.2.1.

4.3.8 *Odor.*—Note the odor of the sample immediately upon opening the container.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.—Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Glass bottles.* — When specified by the contract or order (see 6.2), one kilogram of triphenyl phosphate shall be packaged in a screw cap, amber colored bottle conforming to type I, class A, of Federal Specification U–M–186. The bottle shall be sealed with a secondary closure of pressure sensitive tape conforming to Military Specification MIL–T–4239. The tape shall be of sufficient width to extend from the top of the cap to well down over the bottle neck and shall be one and one third the cap circumference in length. The bottle shall be enclosed in a telescoping type water-resistant fiber can conforming to type II of Military Specification MIL–C–3955 and shall be cushioned with either vermiculite or shredded asbestos not less than three-fourths inch thick. The fiber cans shall be sealed with pressure sensitive tape conforming to type I, class 2 of Federal Specification PPP–T–60.

5.1.1.1.1 *Intermediate packaging.*—Sixteen of the fiber can-enclosed glass bottles shall be intermediately packaged in a fiber box conforming to type I, class 2, style RSC, of Federal Specification PPP–B–636. The box shall be closed in accordance with the appendix to the box specification.

5.1.1.2 *Metal cans.*—When specified by the contract or order (see 6.2), triphenyl phosphate shall be packaged in metal cans in 2-pound or 10-pound quantities. Metal cans shall conform to type V, class 4, finish plan B, with side seam striping, of Federal Specification PPP–C–96.

5.1.1.3 *Metal drums.* — When specified in the contract or order (see 6.2) triphenyl phosphate shall be packaged in metal drums in quantities of 50 or 550 pounds. 50-pound capacity drums shall conform to type I, class 1, of Federal Specification PPP–D–760 and 550-pound capacity drums shall conform to type II of Federal Specification PPP–D–729.

5.1.1.4 *Bulk.*—When specified in the contract or order (see 6.2) triphenyl phosphate shall be shipped bulk.

5.1.2 *Level C.*—Unless otherwise specified triphenyl phosphate shall be packaged in containers conforming to the Uniform Freight Classification Rules or other regulations applicable to such supplies.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Glass bottles.* — Four intermediate packages, packaged as specified in 5.1.1.1.1 shall be packed in a fiber box conforming to type I, W5C, style RSC of Federal Specification PPP–B–636. The box shall be closed and strapped in accordance with the appendix to the box specification.

5.2.1.2 *Metal cans.*—Unless otherwise specified, twelve 2-pound, or six 10-pound cans

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shall be packed in a fiber box conforming to type I, W5C, style RSC, of Federal Specification PPP–B–636. The box shall be closed and strapped in accordance with the appendix to the box specification.

5.2.1.3 *Metal drums.*—Fifty-pound and 550-pound drums require no overpacking.

5.2.2 *Level B.*

5.2.2.1 *Glass bottles.*—Glass bottles shall be packed as specified in 5.2.1.1 except that the fiber box shall conform to type I, class 1 of Federal Specification PPP–B–636.

5.2.2.2 *Metal cans.*—Metal cans shall be packed as specified in 5.2.1.2 except that the fiber box shall conform to type I, class 1 of Federal Specification PPP–B–636.

5.2.3 *Level C.*—Unless otherwise specified triphenyl phosphate shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall comply with the Uniform Freight Classification Rules and other regulations applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civilian agencies.*—Marking for civilian agencies shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special marking required by the contract or order, containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.—This specification covers triphenyl phosphate, a reaction product of phenol and a phosphorous compound such as phosphorous oxychloride or a phosphorous halide. It is intended for use in the manufacture of organic coatings.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Level of packaging and packing required (see 5.1 and 5.2).

(*c*) Size of package and quantity required.

6.3 Triphenyl phosphate should be purchased by weight, the unit being a U. S. pound.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C E M O Q
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Copies of this specification may be purchased f[illegible] s each.

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